Crowdsourced Civic Issue Reporting and Resolution System

A project report

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ABSTRACT

Urban environments continue to grow rapidly, and with this expansion comes a wide range of civic challenges related to infrastructure maintenance, sanitation, public safety, and general urban upkeep. Municipal authorities often struggle to receive timely and accurate information about issues on the ground, largely due to the limitations of traditional reporting mechanisms such as manual complaint registration, telephone-based grievance systems, and periodic inspections. These outdated processes frequently result in delayed responses, unresolved issues, and a lack of transparency for citizens.

Civic Connect is developed as a modern, technology-driven solution to bridge this gap by creating an accessible, citizen-centric platform for reporting civic issues in real time. The system integrates a user-friendly Android mobile application with a robust cloud backend and an artificial intelligence–powered automation pipeline. Citizens can quickly capture images of civic problems, such as potholes, water leaks, garbage piles, or malfunctioning streetlights, and submit them instantly along with location data and descriptions. These reports are then processed through an automated workflow where AI classifiers identify the issue type, while a duplicate detection engine evaluates visual and spatial similarity to avoid redundant submissions.

The project also incorporates a dedicated web-based administrative dashboard that enables municipal staff to view, filter, manage, and respond to reported issues efficiently. By combining real-time updates, automated categorization, and transparent tracking, Civic Connect strengthens communication between citizens and administration, enhancing public participation and accountability.

This report presents a comprehensive study of the system's design, requirements, architecture, workflows, and development methodology. It also outlines the motivation for the project, the problem domain, and the technologies employed throughout implementation. Civic Connect demonstrates how mobile technology and AI can be integrated to streamline civic problem reporting, ensure faster decision-making, reduce manual effort, and ultimately contribute to the creation of smarter, more responsive urban communities.

INTRODUCTION

Objective of the Project

The primary objective of the *Civic Connect* project is to design and develop an efficient, transparent, and user-friendly digital platform that enables citizens to easily report civic infrastructure issues and allows municipal authorities to respond to these issues more effectively. The project aims to modernize the existing civic complaint system by integrating mobile technologies, cloud services, and artificial intelligence to streamline the entire reporting and resolution workflow.

A major objective is to empower citizens by providing them with a simple mobile application through which they can capture images of civic problems, automatically attach GPS location data, and submit concise descriptions. This shifts civic reporting from a slow, manual process to a fast, accessible, and participatory model. By removing barriers to reporting, the system encourages greater community involvement and ensures that issues affecting public safety or infrastructure quality are addressed promptly.

Another core objective is the automation of early-stage administrative tasks that traditionally consume significant time and manpower. Through AI-based classification, the system automatically categorizes reported issues, reducing the burden on municipal staff to manually review each image. Additionally, duplicate detection ensures that multiple reports of the same issue are identified and consolidated, improving data consistency and preventing redundant work.

The project also aims to create a structured and transparent workflow for authorities. This includes providing a centralized administrative dashboard where staff can view all reports, filter them by category or priority, update their statuses, and track resolution progress. Such a system enhances accountability, improves response times, and ensures systematic follow-up.

Overall, the objective of Civic Connect is not only to build a technological platform, but also to foster a more connected, responsive, and sustainable urban environment by strengthening collaboration between citizens and governing bodies. Through real-time reporting, automated

processing, and transparent monitoring, the project contributes to smarter city management and improved civic engagement.

Description of the Project

The *Civic Connect* project is designed as a comprehensive digital platform that simplifies the way civic issues are reported, tracked, and managed. It brings together a mobile application for citizens, a cloud-based backend for secure data processing, and an administrative dashboard for municipal authorities. The goal is to create an end-to-end solution that supports real-time reporting, intelligent categorization, and transparent resolution of civic problems.

At its core, the system revolves around a user-friendly Android mobile application. This application enables citizens to capture images of civic issues—such as potholes, waste accumulation, broken streetlights, water leakage, or damaged public infrastructure—and submit them instantly. The app automatically captures GPS coordinates, ensuring that the location of the issue is accurately recorded. Users can also enter a short description to provide additional context. The interface is intentionally kept simple to ensure accessibility for users of different age groups and technological backgrounds.

Once a report is submitted, it is processed by a cloud-backed infrastructure powered by Firebase. Issue metadata is stored in Firestore, while images are securely uploaded to Firebase Storage. This integration ensures that the data is synchronized in real-time and can be accessed instantly by municipal administrators. Firebase Authentication manages secure user registration and login, preventing unauthorized access and protecting sensitive data.

A distinguishing feature of the project is the integration of artificial intelligence. A trained image classification model automatically analyzes the submitted image and predicts the issue category. This eliminates the need for manual sorting by administrative staff and speeds up the triaging process. In addition, a duplicate detection engine evaluates both the visual similarity of the image and its geographic proximity to previously reported issues. If a similar issue already exists, the system flags the report as a duplicate, reducing redundancy and helping administrators focus on unique, unresolved cases.

An administrative dashboard forms the management layer of the system. It provides municipal authorities with tools to view all reports, filter them based on category, location, or severity, and update their statuses. Administrators can change the progress of an issue from “Pending” to “In Progress” or “Resolved,” ensuring transparency and accountability. The dashboard also helps identify common problem areas within a locality and supports data-driven decision-making.

Overall, the Civic Connect project offers a complete digital transformation of the civic issue reporting process. By combining citizen participation, cloud computing, and AI automation, the system enhances the efficiency of urban maintenance operations and promotes a more responsive and connected municipal governance model.

Scope of the Project

The scope of the *Civic Connect* project focuses on developing a functional, scalable, and user-centered system that addresses the essential needs of civic issue reporting and management. The project is intentionally structured to deliver maximum value within a manageable technological and operational boundary, making it suitable for deployment in municipalities, campus environments, residential communities, or smart-city pilot initiatives.

At the citizen level, the scope includes the development of an intuitive Android mobile application that enables users to report civic issues quickly and effortlessly. The core features of the mobile app encompass capturing or uploading images, automatic GPS location tagging, entering short descriptions, and viewing or upvoting issues reported by others. These features ensure that citizens can easily participate in identifying civic problems within their surroundings.

On the backend side, the project includes the full integration of Firebase services, such as Firestore for real-time issue storage, Firebase Storage for image management, and Firebase Authentication for secure user access. The backend is responsible for maintaining fast synchronization, ensuring data integrity, and supporting high availability for simultaneous users across different geographical areas.

A significant part of the project's scope involves the implementation of artificial intelligence to automate critical administrative processes. The AI module is designed to classify reported issues into predefined categories and detect duplicates using both visual similarity and geographic proximity. This automated behavior reduces manual intervention, speeds up verification, and ensures consistent classification across all submissions.

The administrative dashboard forms another essential component within the project's scope. It provides municipal staff or authorized authorities with a user-friendly interface to view, monitor, filter, and resolve issues. Through this dashboard, administrators can update issue statuses, access detailed information, and analyze commonly reported problem areas. The dashboard thus acts as the command center for managing and closing the feedback loop between citizens and authorities.

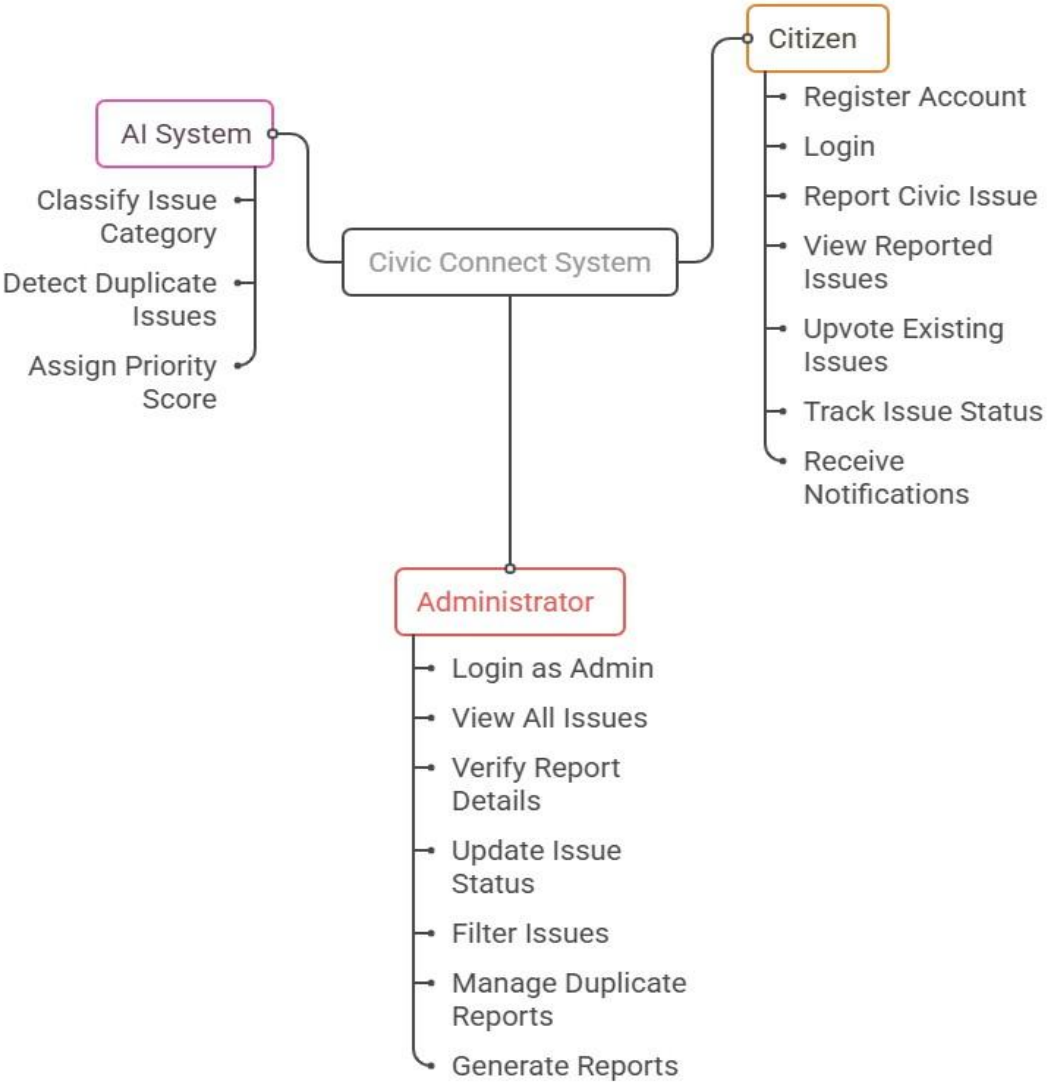
The scope of the project is also limited in certain areas to maintain focus and feasibility. For instance, integrations with official municipal ERP systems, automated field-worker allocation, advanced predictive analytics, multi-language support, and offline-first functionality are not included in this phase. While these features offer significant value, they are considered part of future expansions and not within the expected deliverables of this capstone project.

In summary, the scope of *Civic Connect* covers the full development of a real-time reporting platform using mobile, cloud, and AI technologies, while maintaining clear boundaries that allow the project to remain practical, achievable, and effective. It lays the groundwork for a scalable solution that can be expanded with more advanced features in future iterations.

Use Case Model

The Use Case Model for the *Civic Connect* system provides a structured representation of the functional interactions between the system and its primary actors. It visually illustrates how different users engage with the system and outlines the major services or functionalities offered. The model serves as the starting point for understanding the system's overall behavior, ensuring that all stakeholder requirements are accurately captured and mapped before moving into design and implementation.

Civic Connect System Use Case Diagram



SYSTEM DESCRIPTION

Customer/User Profiles

The effectiveness of the *Civic Connect* system depends heavily on understanding the different types of users who interact with it. Each user group has distinct goals, expectations, and responsibilities, and the system is designed to support their needs through tailored functionalities. The primary user profiles include Citizens, Administrators, and certain Automated System Components that operate as internal actors within the platform.

1. Citizens

Citizens represent the largest and most essential group of users within the Civic Connect ecosystem. They are everyday individuals living within the community who encounter civic issues directly in their local surroundings. The system is designed to empower them by providing a simple and accessible way to report problems in real time.

Citizens typically rely on mobile devices to capture images of issues such as potholes, overflowing garbage bins, blocked drainage, water leakages, or malfunctioning streetlights. For this reason, the mobile application interface is intentionally kept clean and intuitive to support users who may have varying levels of technical proficiency. Critical features such as automatic GPS location capture, pre-filled metadata, and quick submission workflows minimize effort and encourage participation.

Beyond reporting, citizens also have the option to upvote issues reported by others. This feature helps highlight problems that affect a larger portion of the community, allowing administrators to prioritize them when necessary. Citizens can also track the status of their own submissions, giving them visibility into municipal efforts and building a sense of transparency and trust.

2. Administrators (Municipal Staff)

Administrators are official representatives of municipal bodies, local government units, or authorized maintenance personnel responsible for overseeing civic infrastructure. They form the second major user group and are given elevated access within the system.

Administrators log in through a dedicated dashboard where they can review reported issues in real time. Their responsibilities include verifying the authenticity of reports, examining the descriptions and images, and updating issue statuses based on field inspections or work progress. They can categorize issues, mark them as “In Progress,” and finally “Resolved” once the problem has been addressed.

The system also enables administrators to filter issues by category, location, severity, or submission date, allowing them to efficiently manage large volumes of reports. Through this dashboard, administrators gain insights into recurring problem areas, allowing them to plan maintenance work more effectively. Ultimately, the administrator profile is designed to streamline municipal workflows, reduce delays, and ensure that civic resources are deployed efficiently.

3. Automated System Components (AI Actor)

While not a human user, the *AI System* is considered an important internal actor within Civic Connect. It performs several automated tasks that directly influence how reports are processed and prioritized.

The AI module analyses submitted images to classify the issue into categories such as pothole, waste, water leak, or streetlight issue. This eliminates the need for manual sorting and ensures consistent classification. Additionally, the duplicate detection component compares new submissions with existing ones using visual similarity metrics and proximity checks. If a submission is considered a duplicate, the system flags it for administrative review, reducing redundancy and minimizing manual workload.

The AI actor enhances the system's responsiveness, ensures uniform processing of reports, and supports administrators by handling repetitive tasks with high accuracy and speed.

Summary

Together, these user profiles form the backbone of the Civic Connect platform. Citizens supply real-world data, administrators act on the information, and the AI system automates key processes to maintain efficiency. This combination creates a balanced ecosystem that improves civic engagement, accelerates issue resolution, and enhances overall urban governance.

Assumptions and Dependencies

Assumptions

- Users have access to an Android smartphone with a working camera and GPS.
- Users have an active internet connection while submitting reports.
- Citizens will provide accurate information and clear images when reporting issues.
- Municipal authorities will regularly monitor the dashboard and update issue statuses.
- AI models will perform reliably under typical image conditions and lighting.

Dependencies

- **Firebase Services** (Authentication, Firestore, Storage) must remain operational for the system to function.
- **Google Play Services** are required for accurate location detection.
- **TensorFlow/TFLite** models must be available for classification and duplicate detection.
- **Web browsers** used by administrators must support modern dashboard features.
- The system relies on **stable cloud infrastructure** to ensure real-time updates and data synchronization.

Functional Requirements

1. **User Registration and Login:**

The system must allow citizens and administrators to create accounts and log in securely.

2. **Issue Reporting:**

Users must be able to submit civic issues with an image, GPS location, and short description.

3. **Image Upload and Storage:**

The system must upload issue images to cloud storage and link them to issue records.

4. **AI Classification:**

The system must automatically classify issues into predefined categories based on the uploaded image.

5. **Duplicate Detection:**

The system must detect and flag duplicate issue reports using image similarity and location data.

6. **View and Upvote Issues:**

Citizens must be able to view reported issues and upvote those they find relevant.

7. **Admin Dashboard:**

Administrators must be able to view all issues, filter them, and update issue statuses (Pending, In Progress, Resolved).

8. **Real-Time Synchronization:**

Any update made by users or administrators must be reflected in the system in real time.

Non-Functional Requirements

1. **Performance:**

The system should load pages quickly and process reports (upload + classification) within a few seconds.

2. **Scalability:**

The backend should support increasing numbers of users and issue submissions without performance loss.

3. **Security:**

User data must be protected through authentication, access control, and secure cloud storage.

4. **Usability:**

The mobile app and dashboard must be simple, intuitive, and accessible to non-technical users.

5. **Reliability:**

The system must maintain consistent performance and accurate real-time data synchronization.

6. **Availability:**

Cloud services must remain available with minimal downtime to ensure uninterrupted reporting.

7. **Maintainability:**

The system's codebase and architecture must support easy updates, bug fixes, and future enhancements.

DESIGN

System Design

The system design of *Civic Connect* outlines how different components of the platform interact with one another to support seamless reporting, processing, and management of civic issues. It serves as the architectural blueprint that defines the system's structure, data flow, and the coordination of major modules. The design ensures that the application is efficient, scalable, and capable of handling real-time updates while providing a smooth user experience for both citizens and administrators.

The system follows a multi-layered architecture comprising the mobile client layer, backend processing layer, AI automation layer, and administrative dashboard interface. Each layer performs a specific set of operations, yet they work together cohesively to achieve the overall

purpose of civic issue reporting and resolution.

The mobile client layer is responsible for gathering input from the user. This includes capturing or uploading an image of the issue, automatically retrieving the GPS location, and submitting a short textual description. The client communicates with the backend through secure connections and ensures that user inputs are packaged correctly before being sent to the cloud for processing. The mobile app also displays reported issues and status updates, providing a complete user feedback loop.

The backend layer, built on Firebase services, acts as the central processing hub of the system. It stores user data, issue details, and images in Firestore and Firebase Storage. The backend is designed to be event-driven, meaning that certain actions—such as the creation of a new issue—automatically trigger additional processes. Real-time capabilities of Firestore enable instant synchronization across user devices and administrator dashboards.

The AI processing layer adds intelligence and automation to the system. When a new issue is submitted, cloud functions retrieve the image and run it through the classification model to determine the appropriate category. The same module generates an image embedding vector that is used to evaluate similarity to previously reported issues. This ensures that duplicate issues are identified early and flagged accordingly. AI components reduce repetitive manual work and help administrators prioritize their efforts more effectively.

The administrative dashboard layer provides the interface through which municipal staff manage incoming reports. It allows administrators to view all issues, filter them based on category or status, update their progress, and review duplicate flags generated by the AI. The dashboard plays a vital role in maintaining transparency, as administrators' actions are immediately reflected on the citizens' applications through real-time updates.

Together, these design components create a system that is robust, maintainable, and responsive. The architecture ensures that users can report issues effortlessly, while administrators receive accurate, filtered, and prioritized data to support timely interventions. This systematic approach to design helps Civic Connect achieve its goal of improving the efficiency of civic issue reporting and enhancing community engagement.

Entity Relationship Diagram

The Entity–Relationship (ER) diagram for the *Civic Connect* system illustrates the core data entities and the relationships between them. The diagram highlights how information flows within the platform and how different components of the system interact at the data level.

At the center of the diagram is the Issue entity, which represents civic problems reported by

citizens. Each issue is created by a User, indicating a one-to-many relationship where a single user can report multiple issues. The Issue entity is connected to several supporting entities that enhance the system's functionality.

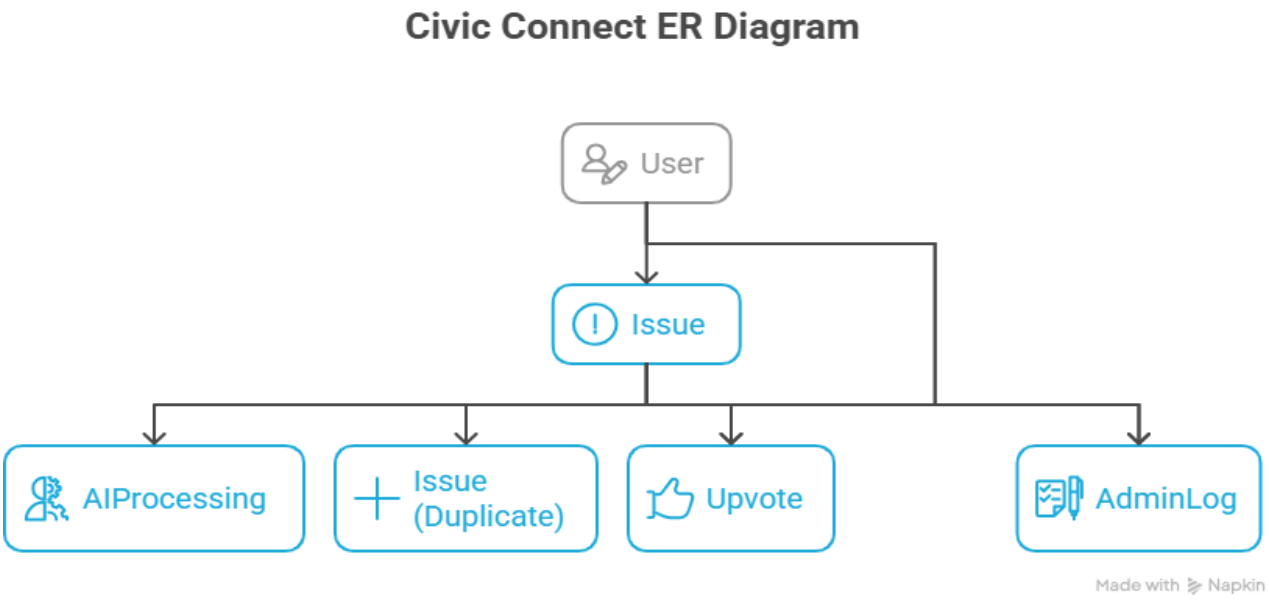
The AIProcessing entity captures the automated classification results and confidence scores generated by the AI model for each reported issue. This creates a one-to-one or optional relationship between Issue and AIProcessing, as each issue may undergo AI processing once it is submitted.

The diagram also includes a Duplicate Issue relationship, represented by a recursive connection from Issue back to itself. This indicates that some issues may be flagged as duplicates of previously reported ones based on visual similarity or location similarity.

The Upvote entity captures citizen engagement. Users can upvote issues reported by others, helping administrators prioritize widely recognized or high-impact problems. This represents an indirect many-to-many relationship between User and Issue, simplified here by showing Upvote directly connected to Issue.

Finally, the AdminLog entity records administrative actions performed on each issue. Administrators can update statuses, verify details, or resolve issues, and each of these actions is logged for transparency and accountability. This forms a one-to-many relationship between an Issue and AdminLog entries.

Overall, the ER diagram provides a structured view of how user data, issue reports, AI results, administrative actions, and community feedback are connected within the Civic Connect system.



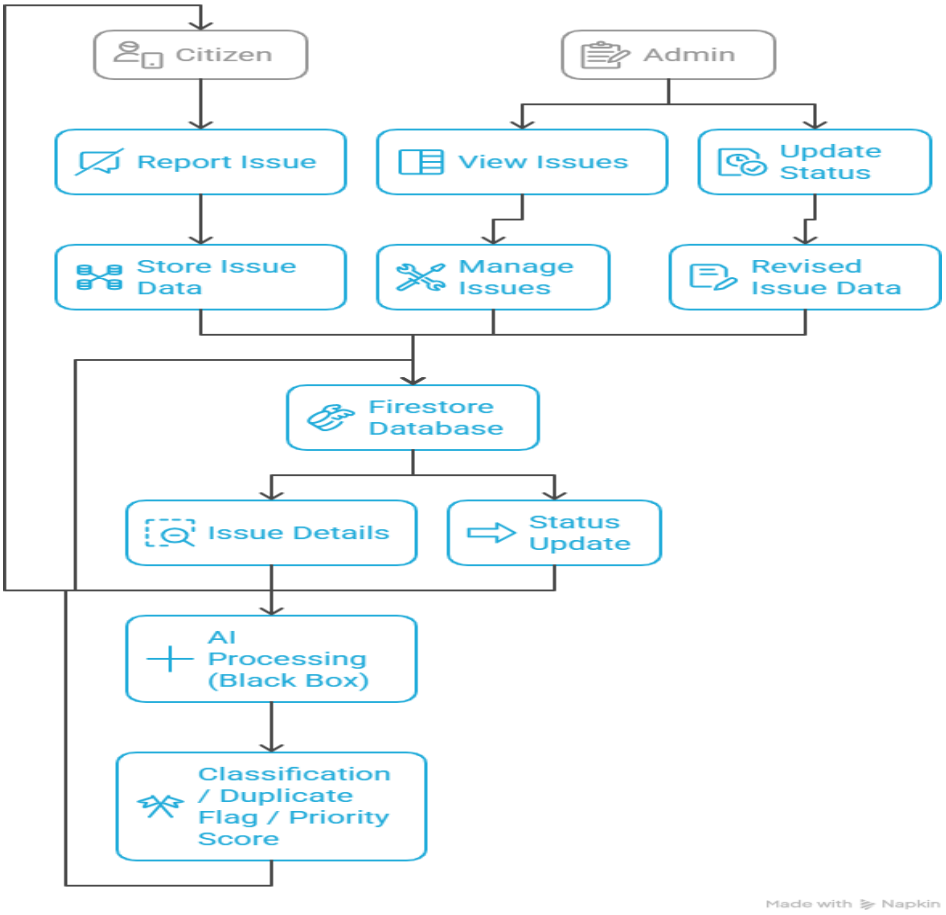
Data Flow Diagram

The Level 1 Data Flow Diagram provides a detailed breakdown of how information moves through the *Civic Connect* system. It expands the major processes into smaller, more specific actions carried out by both citizens and administrators. The diagram begins with the citizen reporting an issue, which includes capturing data such as images, descriptions, and GPS location. This information is then stored in the Firestore Database.

Administrators interact with the system by viewing reported issues, updating their statuses, and managing issue information. Updated or revised data is also stored back in the Firestore Database. The AI Processing module retrieves issue details from the database and performs automated tasks such as image classification, duplicate detection, and priority scoring. The results are then sent back into the system to assist administrators in decision-making.

This DFD highlights how different modules collaborate: citizens submit issues, administrators manage them, the database stores the information, and the AI system enhances accuracy and efficiency. It provides a complete end-to-end view of how raw issue data becomes classified, processed, and actionable within the platform.

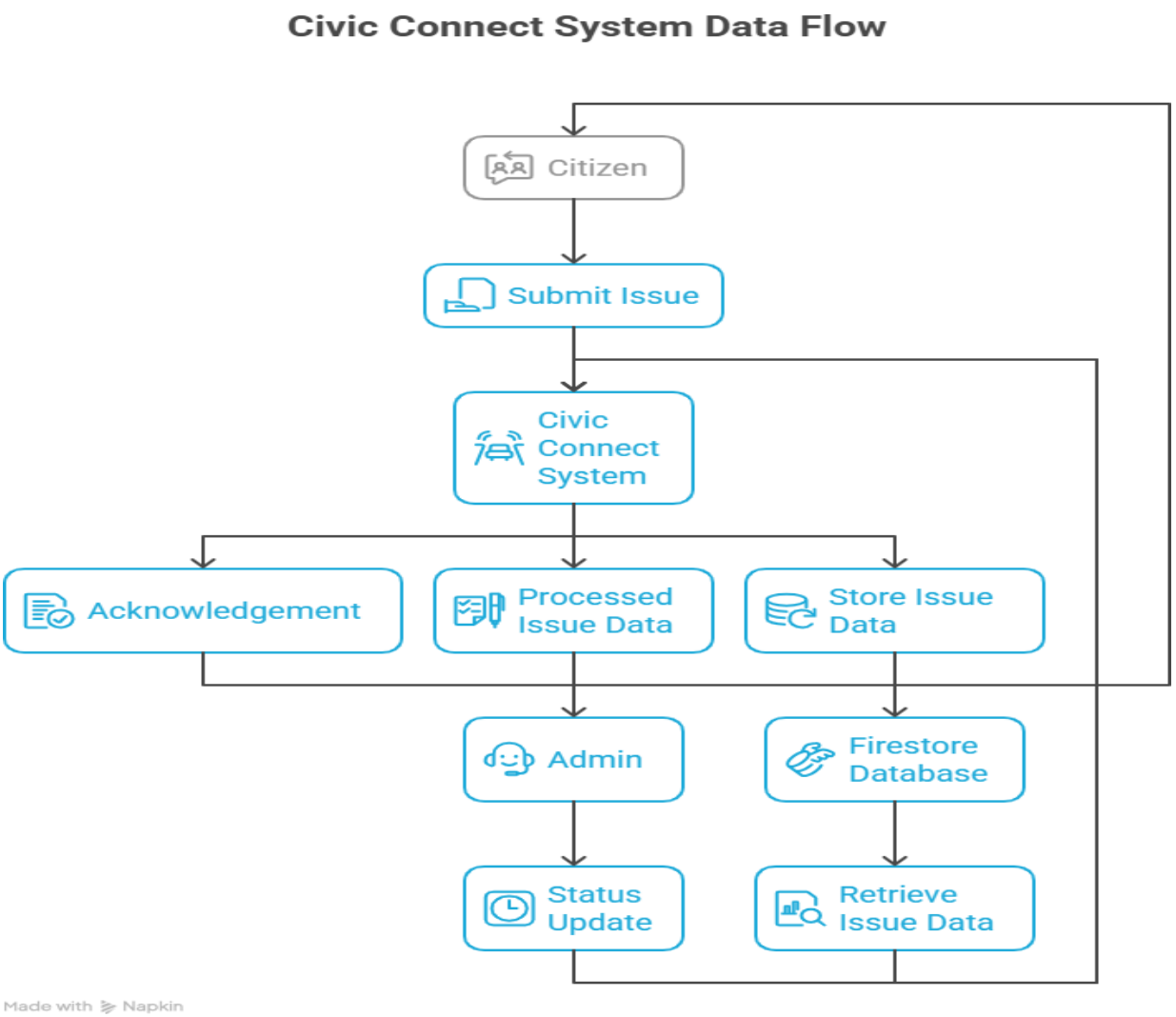
Civic Connect System Data Flow Diagram



The Level 0 Data Flow Diagram represents a simplified, high-level view of the Civic Connect system. It shows the system as a single major process that interacts with external entities such as the citizen and the administrator.

The diagram begins with the citizen submitting an issue. This data flows into the Civic Connect System, which performs internal processing and generates outputs such as acknowledgement messages, processed issue data, and stored issue information. The Firestore Database acts as the core storage location for all submitted issues. Administrators access this processed data to review reports and update their statuses. Any status updates made by administrators flow back into the system and are relayed to the citizen.

This high-level diagram focuses on the overall functionality of the system without breaking down internal components. It demonstrates how information moves among the citizen, the system, the database, and the administrator, showing the fundamental data exchange cycle within Civic Connect.



Database Design

The database design of the *Civic Connect* system is structured to support efficient storage, retrieval, and processing of civic issue reports submitted by users. Since the platform relies heavily on real-time updates, media uploads, and interactions between different user groups, the database architecture must be flexible, scalable, and optimized for fast read–write operations. To meet these needs, the system uses Google Firebase Firestore, a cloud-based NoSQL database, along with Firebase Storage for handling images.

Firestore's document-based design is particularly suitable for this application because issue reports vary in detail, may include additional fields over time, and require rapid synchronization between mobile clients and administrative dashboards. The database design ensures that each entity is mapped to a collection with clearly defined fields, while relationships between entities are managed through document references, foreign keys, and ID-based linking.

The central component of the database is the Issue collection, where each document represents a single civic issue reported by a user. Each issue document contains details such as the user ID of the reporter, the description of the problem, the uploaded image URL, the system-assigned category, geographic coordinates, current status, timestamps, and the number of upvotes it has received. Additional optional fields, such as duplicate markers and AI processing results, allow the system to scale and incorporate more advanced features without restructuring the schema.

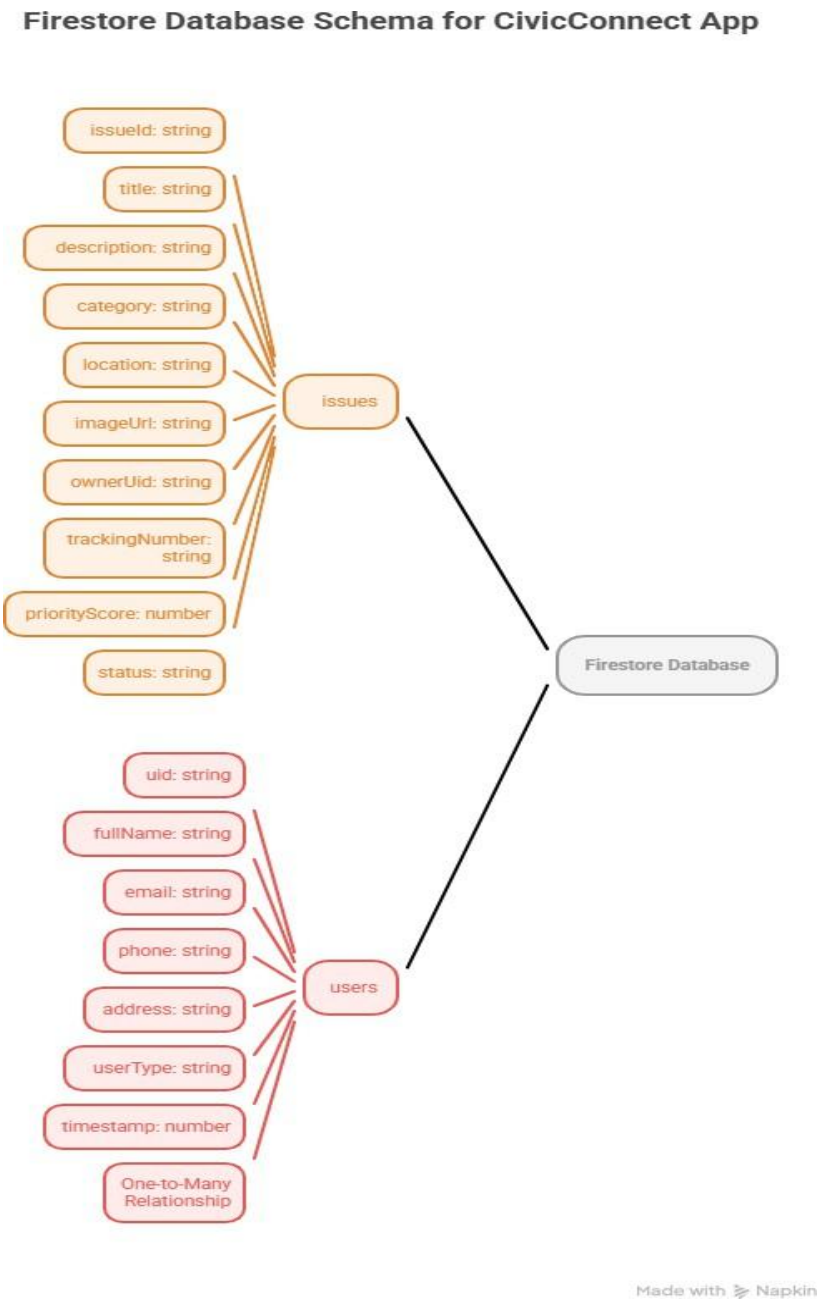
The User collection stores essential information about citizens and administrators. Each user document includes identification details and role attributes to differentiate between regular users and municipal staff. This supports secure access control and role-based actions within the platform.

The Upvote collection is used to track citizen engagement. It acts as a bridge between users and issues, representing a many-to-many relationship: multiple users can upvote multiple issues. Each upvote document records the user who performed the action, the issue being upvoted, and the timestamp.

Administrative actions are captured through the AdminLog collection, which maintains records of all updates made by municipal staff. Logging information such as the type of action, previous and new status, the admin's ID, and timestamp provides transparency and a complete audit trail of issue resolution progress.

Finally, AI-related metadata is handled in the AIProcessing collection, which stores automated classification results, confidence scores, and embedding vectors generated for

duplicate detection. Maintaining this information separately ensures that the system remains modular and that AI components can evolve independently of the core database design. Overall, the database design focuses on simplicity, extensibility, and real-time efficiency. Firestore's NoSQL structure supports dynamic data models, while carefully organized collections and document references ensure that the Civic Connect system can scale smoothly and handle increasing amounts of user-generated data as the platform grows.



SCHEDULING AND ESTIMATES

The development of the *Civic Connect* system follows a structured and phased schedule designed to ensure timely progress and balanced workload distribution across all project components. The scheduling process divides the project into logical phases such as requirement analysis, system design, development, integration, testing, and documentation. Each phase is assigned an estimated duration based on its complexity, dependencies, and the expected effort required from the development team. The schedule takes into consideration the sequential nature of software development activities. For example, requirements must be clearly defined before design begins, and system design must be completed before implementation can start. Similarly, AI model preparation requires a substantial dataset and therefore overlaps with parts of the design and early development stages. By organizing the project in this manner, the team can maintain focus, avoid bottlenecks, and ensure that work progresses smoothly.

The estimates for each phase are based on typical timelines for academic-level software projects involving mobile development, cloud integration, and machine learning components. These estimates account for time required to collect and preprocess datasets, train classification models, set up the backend infrastructure, and build the Android interface. Additional time is allocated for testing and debugging, which are essential for ensuring system reliability and usability.

Overall, the scheduling and estimation plan ensures that the team can deliver a functional, stable, and user-ready system within the planned timeframe. It also provides clear expectations for the workload and milestones, allowing adaptive planning if unexpected challenges arise.

Project Timeline

The project is divided into the following phases:

- **Week 1–2: Requirement Analysis**
  Identify system needs, define user roles, gather functional and non-functional requirements, and prepare initial documentation.
- **Week 3–4: System Design**
  Develop UML diagrams, ER diagrams, Data Flow Diagrams, system architecture layout, and finalize database schema.
- **Week 5–7: Dataset Preparation and AI Model Training**
  Collect relevant images, preprocess data, train the classification model, and validate its performance.
- **Week 6–11: Android Application Development**
  Build the UI screens, integrate camera and GPS, implement issue submission, and connect app to backend services.
- **Week 8–10: Backend and AI Integration**
  Deploy Cloud Functions, connect the AI inference pipeline, and implement real-time data synchronization in Firestore.
- **Week 10–12: Admin Dashboard Development**
  Create the web interface for issue management, filtering options, and status updates.
- **Week 12–14: Testing and Debugging**
  Conduct unit testing, integration testing, and final system testing to ensure correct functionality and usability.
- **Week 15: Deployment and Documentation**
  Finalize the project report, prepare deployment configurations, and complete the handover documentation.

CONCLUSION

The *Civic Connect* system demonstrates how modern mobile technology, cloud computing, and artificial intelligence can be combined to address persistent challenges in civic issue reporting and management. Traditional reporting mechanisms are often slow, inefficient, and lack transparency. By contrast, Civic Connect enables citizens to submit real-time reports with images and location data, allowing municipal authorities to receive immediate, accurate, and actionable information.

Through its multi-layered design, the system ensures seamless data flow—from user submission to backend processing and administrative review. The integration of AI improves overall efficiency by automatically classifying issues and detecting duplicate reports, helping authorities prioritize work and reduce redundancy. The administrative dashboard further enhances operational visibility, enabling municipal staff to manage issues more effectively.

Overall, Civic Connect contributes to a more responsive governance model by encouraging citizen participation, improving communication between the public and authorities, and promoting accountability. The system successfully demonstrates a scalable and practical approach to smart city infrastructure management, making it a valuable model for real-world implementation.

FUTURE SCOPE

While the current version of Civic Connect provides a strong foundation, several enhancements can significantly improve scalability, usability, and municipal integration. One major area of expansion is the use of predictive analytics to anticipate recurring civic problems, allowing authorities to take preventive action rather than responding reactively. Integrating GIS-based heatmaps, real-time dashboards, and trend analysis could further strengthen data-driven decision making.

The system can also evolve through integration with municipal ERP and workflow systems, enabling automatic task creation, field staff assignment, and end-to-end resolution tracking. Expanding multi-language support, voice-assisted reporting, and improved accessibility features would make the app usable for a wider demographic, including elderly users and individuals with disabilities.

From a technical perspective, the AI models can be enhanced using larger datasets and on-device inference to reduce latency. The platform could additionally support offline reporting, SMS-based submissions for low-connectivity areas, and community engagement features such as reward points for active contributors. Implementing geofencing alerts, which notify authorities when multiple issues arise in the same region, can further improve response efficiency.

Overall, these enhancements would make Civic Connect more robust, inclusive, and aligned with the needs of modern smart cities, enabling it to serve as a long-term solution for urban infrastructure management.

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